Supplementary material for

Patient-level differential privacy, membership leakage and explanation fidelity in federated brain tumour MRI classification on a deduplicated benchmark

Authors to be completed

Overview

This supplement reports every configuration behind the main text, together with the controls and selection steps that the article summarises in a sentence. All tables were generated by `scripts/make_supplement.py` from the files frozen at repository tag `v0.9.1-results-frozen`, so each value can be traced to a hashed table or to a run of a frozen grid. Seeds 1 to 5 give every reported mean, seed 0 serves only for selection, and intervals are two-sided 95% t intervals over seeds unless a caption states otherwise.

The sections follow the order of the article. Section S1 documents the duplicate check that defines the clean test set, Section S2 compares the full and the clean test sets and adds the source controls, Section S3 gives the privacy accounting and the selection costs, Section S4 covers client count and drift correction, Section S5 adds the membership-inference detail and Section S6 the explanation detail.

S1 Benchmark curation and near-duplicate check

Perceptual hashing removed exact and near-exact copies during curation, yet it compares images only in one orientation. A second check therefore computed the normalised cross-correlation (NCC) between every pair of curated images at 64×64 pixels, once as stored and once with one image mirrored, keeping the larger value. Part A (Table S1) counts pairs above three thresholds and breaks the cross-split pairs down by the splits they join.

Table S1: Part A of the NCC check. Pairs of curated images (64×64) at or above each normalised cross-correlation threshold, taking the larger of the plain and the mirrored comparison.

NCC $\geq$	Pairs	Same unit	Cross-split	Test–train	Test–val	Train–val	Mirror only	Label conflict
0.9	2296	1665	295	190	29	76	139	19
0.95	682	357	148	84	17	47	120	0
0.98	195	28	80	43	9	28	79	0

Across the 813 test images, the NCC to the nearest image in another split has median 0.809, 90th percentile 0.926 and maximum 1.0000. 66 test images reach 0.95 and form the contaminated set.

Table S2: Part B of the NCC check. Agreement between perceptual-hash duplicates removed during curation and their kept partner, by Hamming distance d of the 64-bit hash.

d	Pairs	Median NCC	Min NCC	< 0.90	< 0.95	< 0.98
0	2945	1.0000	0.657	6	10	17
2	733	0.9992	0.629	13	22	82
4	296	0.9735	0.558	80	93	176

At the threshold of 0.95, fixed before the check, 148 pairs crossed a split boundary and 120 of them matched only after mirroring. Part B (Table S2) confirms that the hash and the correlation agree on the duplicates that curation removed, which shows that the correlation threshold is not looser than the hash it audits. Figure S1 shows the most similar cross-split pairs, most of them Br35H no-tumour images whose mirrored twin sat in another split.

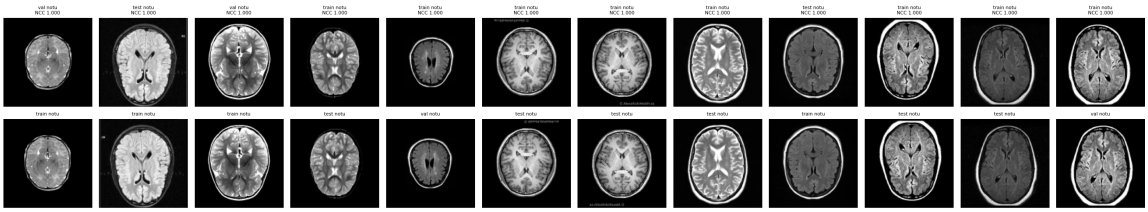


Figure S1: Cross-split pairs with the highest NCC after mirroring. Each pair shows the test image and its nearest image in another split. These pairs define the 66 contaminated test images excluded from every result in the article.

S2 Full and clean test sets, and source controls

Table S3 lists balanced accuracy for every reported configuration on the full and on the clean test set. Removing the contaminated images lowered accuracy in every configuration, mostly because no-tumour sensitivity fell once the mirrored Br35H twins left the test set, yet no ranking between settings changed.

Table S3: Balanced accuracy on the full test set (813 images) and on the clean test set (747 images) for every reported configuration, mean over seeds 1 to 5 with 95% t interval. sens$_N$ is the no-tumour sensitivity.

Training	Algorithm	Regime	lr	Bal. acc. (full)	Bal. acc. (clean)	sens$_N$ full	sens$_N$ clean
central	–	no DP	0.02	0.919 (0.907, 0.931)	0.908 (0.894, 0.921)	0.958	0.913
central	–	clip only	0.02	0.871 (0.850, 0.891)	0.845 (0.818, 0.871)	0.902	0.800
central	–	$\varepsilon = 8$	0.02	0.815 (0.779, 0.851)	0.778 (0.739, 0.818)	0.846	0.700
central	–	$\varepsilon = 3$	0.02	0.686 (0.616, 0.756)	0.648 (0.572, 0.723)	0.738	0.587
central	–	$\varepsilon = 1$	0.02	0.331 (0.269, 0.393)	0.326 (0.272, 0.379)	0.138	0.117
central	–	patient clip only	0.02	0.884 (0.866, 0.903)	0.872 (0.852, 0.891)	0.954	0.903
central	–	patient $\varepsilon = 64$	0.02	0.836 (0.800, 0.873)	0.820 (0.778, 0.862)	0.938	0.873
central	–	patient $\varepsilon = 32$	0.02	0.745 (0.698, 0.792)	0.727 (0.682, 0.771)	0.922	0.850
central	–	patient $\varepsilon = 16$	0.02	0.584 (0.558, 0.611)	0.565 (0.543, 0.588)	0.902	0.827
central	–	patient $\varepsilon = 8$	0.02	0.353 (0.293, 0.413)	0.337 (0.284, 0.390)	0.890	0.823
central	–	patient $\varepsilon = 3$	0.02	0.253 (0.245, 0.261)	0.251 (0.248, 0.253)	0.992	0.983
Dir. 0.1, K=10	FedAvg	no DP	0.01	0.843 (0.834, 0.853)	0.817 (0.806, 0.828)	0.882	0.777
Dir. 0.1, K=10	FedAvg	$\varepsilon = 8$	0.003	0.677 (0.632, 0.722)	0.635 (0.588, 0.682)	0.614	0.447
Dir. 0.1, K=10	FedAvg	$\varepsilon = 3$	0.003	0.632 (0.586, 0.678)	0.589 (0.555, 0.624)	0.498	0.327
Dir. 0.1, K=10	FedAvg	$\varepsilon = 1$	0.003	0.322 (0.303, 0.341)	0.317 (0.301, 0.333)	0.197	0.177
Dir. 0.1, K=10	SCAFFOLD	no DP	0.01	0.815 (0.794, 0.836)	0.790 (0.773, 0.806)	0.901	0.800
Dir. 0.1, K=10	SCAFFOLD	$\varepsilon = 8$	0.003	0.705 (0.682, 0.728)	0.661 (0.632, 0.690)	0.757	0.580
Dir. 0.1, K=10	SCAFFOLD	$\varepsilon = 3$	0.003	0.633 (0.571, 0.696)	0.592 (0.530, 0.653)	0.549	0.383
Dir. 0.5, K=10	FedAvg	no DP	0.01	0.878 (0.863, 0.893)	0.870 (0.852, 0.887)	0.970	0.937
Dir. 0.5, K=10	FedAvg	$\varepsilon = 8$	0.003	0.718 (0.686, 0.750)	0.669 (0.626, 0.712)	0.771	0.577
Dir. 0.5, K=10	FedAvg	$\varepsilon = 8$	0.01	0.662 (0.586, 0.739)	0.624 (0.565, 0.684)	0.528	0.377
Dir. 0.5, K=10	FedAvg	$\varepsilon = 3$	0.003	0.626 (0.570, 0.682)	0.590 (0.548, 0.631)	0.550	0.403
Dir. 0.5, K=10	FedAvg	$\varepsilon = 1$	0.003	0.341 (0.286, 0.396)	0.331 (0.280, 0.383)	0.326	0.287
Dir. 0.5, K=10	SCAFFOLD	no DP	0.01	0.889 (0.871, 0.906)	0.877 (0.857, 0.897)	0.957	0.910
Dir. 0.5, K=10	SCAFFOLD	$\varepsilon = 8$	0.003	0.713 (0.681, 0.744)	0.666 (0.623, 0.708)	0.784	0.597
Dir. 0.5, K=10	SCAFFOLD	$\varepsilon = 3$	0.003	0.621 (0.560, 0.682)	0.582 (0.537, 0.627)	0.558	0.403
Dir. 0.5, K=15	FedAvg	no DP	0.01	0.903 (0.897, 0.909)	0.886 (0.874, 0.899)	0.938	0.870
Dir. 0.5, K=15	FedAvg	$\varepsilon = 8$	0.003	0.696 (0.618, 0.774)	0.655 (0.576, 0.735)	0.723	0.560
Dir. 0.5, K=15	FedAvg	$\varepsilon = 8$	0.01	0.564 (0.464, 0.664)	0.525 (0.449, 0.601)	0.435	0.280
Dir. 0.5, K=15	FedAvg	$\varepsilon = 3$	0.003	0.558 (0.458, 0.658)	0.525 (0.448, 0.602)	0.398	0.267
Dir. 0.5, K=3	FedAvg	no DP	0.01	0.868 (0.850, 0.886)	0.853 (0.827, 0.878)	0.917	0.857
Dir. 0.5, K=3	FedAvg	$\varepsilon = 8$	0.003	0.717 (0.644, 0.790)	0.675 (0.597, 0.754)	0.693	0.527
Dir. 0.5, K=3	FedAvg	$\varepsilon = 8$	0.01	0.820 (0.789, 0.852)	0.779 (0.743, 0.816)	0.794	0.630
Dir. 0.5, K=3	FedAvg	$\varepsilon = 3$	0.003	0.719 (0.698, 0.739)	0.677 (0.644, 0.710)	0.717	0.550
Dir. 0.5, K=5	FedAvg	no DP	0.01	0.906 (0.894, 0.919)	0.892 (0.879, 0.904)	0.946	0.887
Dir. 0.5, K=5	FedAvg	$\varepsilon = 8$	0.003	0.735 (0.712, 0.758)	0.694 (0.667, 0.721)	0.790	0.627
Dir. 0.5, K=5	FedAvg	$\varepsilon = 8$	0.01	0.773 (0.713, 0.833)	0.731 (0.665, 0.798)	0.779	0.613

Training	Algorithm	Regime	lr	Bal. acc. (full)	Bal. acc. (clean)	sens$_N$ full	sens$_N$ clean
Dir. 0.5, K=5	FedAvg	$\varepsilon = 3$	0.003	0.708 (0.684, 0.732)	0.669 (0.644, 0.693)	0.744	0.587
Dir. 5.0, K=10	FedAvg	no DP	0.01	0.899 (0.878, 0.919)	0.884 (0.858, 0.909)	0.946	0.887
Dir. 5.0, K=10	FedAvg	$\varepsilon = 8$	0.003	0.720 (0.695, 0.745)	0.674 (0.636, 0.712)	0.792	0.607
Dir. 5.0, K=10	FedAvg	$\varepsilon = 3$	0.003	0.623 (0.571, 0.676)	0.583 (0.542, 0.625)	0.550	0.390
Dir. 5.0, K=10	FedAvg	$\varepsilon = 1$	0.003	0.333 (0.280, 0.385)	0.328 (0.277, 0.379)	0.266	0.247
Dir. 5.0, K=10	SCAFFOLD	no DP	0.01	0.907 (0.897, 0.917)	0.892 (0.879, 0.905)	0.946	0.887
Dir. 5.0, K=10	SCAFFOLD	$\varepsilon = 8$	0.003	0.716 (0.692, 0.740)	0.669 (0.632, 0.705)	0.789	0.600
Dir. 5.0, K=10	SCAFFOLD	$\varepsilon = 3$	0.003	0.625 (0.563, 0.686)	0.585 (0.535, 0.635)	0.562	0.403
IID, K=10	FedAvg	no DP	0.01	0.904 (0.893, 0.914)	0.891 (0.879, 0.904)	0.954	0.903
IID, K=10	FedAvg	$\varepsilon = 8$	0.003	0.720 (0.698, 0.742)	0.673 (0.641, 0.704)	0.790	0.600
IID, K=10	FedAvg	$\varepsilon = 3$	0.003	0.639 (0.608, 0.671)	0.593 (0.558, 0.627)	0.606	0.420
IID, K=10	FedAvg	$\varepsilon = 1$	0.003	0.348 (0.311, 0.386)	0.344 (0.308, 0.380)	0.304	0.287
IID, K=10	SCAFFOLD	no DP	0.01	0.909 (0.892, 0.926)	0.896 (0.878, 0.914)	0.952	0.900
IID, K=10	SCAFFOLD	$\varepsilon = 8$	0.003	0.719 (0.699, 0.740)	0.671 (0.642, 0.701)	0.789	0.597
IID, K=10	SCAFFOLD	$\varepsilon = 3$	0.003	0.637 (0.605, 0.668)	0.589 (0.555, 0.623)	0.600	0.410
natural3	FedAvg	no DP	0.01	0.875 (0.868, 0.882)	0.841 (0.833, 0.848)	0.824	0.687
natural3	FedAvg	$\varepsilon = 8$	0.01	0.808 (0.783, 0.833)	0.765 (0.731, 0.800)	0.757	0.587
natural3	FedAvg	$\varepsilon = 3$	0.003	0.714 (0.707, 0.721)	0.674 (0.659, 0.688)	0.750	0.590
natural3	FedAvg	$\varepsilon = 1$	0.003	0.537 (0.474, 0.601)	0.525 (0.463, 0.586)	0.366	0.317
natural3	FedAvg	weight noise $s = 0.5$	0.01	0.259 (0.235, 0.284)	0.254 (0.243, 0.266)	0.173	0.153
natural3	FedAvg	weight noise $s = 0.003$	0.01	0.843 (0.826, 0.860)	0.801 (0.785, 0.816)	0.699	0.530
natural3	FedAvg	patient $\varepsilon = 8$	0.003	0.403 (0.333, 0.472)	0.398 (0.330, 0.466)	0.123	0.103
natural3	SCAFFOLD	no DP	0.01	0.900 (0.889, 0.911)	0.879 (0.863, 0.895)	0.920	0.837
natural3	SCAFFOLD	$\varepsilon = 8$	0.01	0.825 (0.801, 0.849)	0.790 (0.757, 0.822)	0.837	0.697
natural3	SCAFFOLD	$\varepsilon = 3$	0.003	0.713 (0.706, 0.719)	0.671 (0.661, 0.681)	0.770	0.603
unit Dir. 0.5, K=10	FedAvg	patient $\varepsilon = 8$	0.01	0.264 (0.234, 0.294)	0.271 (0.235, 0.307)	0.512	0.540

Local-only training (Table S4) shows what each natural client achieves alone. The Br35H client holds no tumour images and cannot exceed the chance level of 0.25, and even the large Figshare client stays far below the federated model, so federation adds real value on this partition.

Table S4: Local-only baselines. Each client of the natural partition trains alone on its own images with the central protocol and is tested on the shared test set (seed means).

Client	Regime	Seeds	Bal. acc. (full)	Bal. acc. (clean)
Br35H	$\varepsilon = 8$	3	0.249	0.255
Br35H	no DP	3	0.250	0.250
Figshare	$\varepsilon = 8$	3	0.464	0.464
Figshare	no DP	3	0.625	0.625
SARTAJ	$\varepsilon = 8$	3	0.245	0.250
SARTAJ	no DP	3	0.543	0.526

The per-source controls in Table S5 separate source composition from diagnostic skill. SARTAJ is the only source with all four classes, and its four-class balanced accuracy stays well below the pooled value in every setting that learns more than chance, which shows how much the pooled number benefits from sources that hold only one or two classes.

Table S5: Per-source controls on the full test set (FedAvg, seed means). SARTAJ is the only source with all four classes, and Figshare holds the three tumour classes only. Specificity is the no-tumour recall within each source.

Mode	Partition	Regime	Bal. acc.	SARTAJ (4 cl.)	Figshare (3 cl.)	Tumour sens.	Spec. SARTAJ	Spec. Br35H
central	natural3	no DP	0.919	0.663	0.924	0.984	0.919	0.994
central	natural3	clip only	0.871	0.586	0.885	0.975	0.797	0.997
central	natural3	$\varepsilon = 8$	0.815	0.528	0.830	0.975	0.702	0.976
central	natural3	$\varepsilon = 3$	0.686	0.467	0.681	0.958	0.590	0.870
central	natural3	$\varepsilon = 1$	0.331	0.326	0.394	0.967	0.125	0.148
central	natural3	patient clip only	0.884	0.636	0.882	0.974	0.902	1.000
central	natural3	patient $\varepsilon = 64$	0.836	0.617	0.818	0.962	0.871	0.997
central	natural3	patient $\varepsilon = 32$	0.745	0.561	0.695	0.960	0.847	0.988
central	natural3	patient $\varepsilon = 16$	0.584	0.449	0.487	0.845	0.820	0.976
central	natural3	patient $\varepsilon = 8$	0.353	0.290	0.177	0.462	0.824	0.948
central	natural3	patient $\varepsilon = 3$	0.253	0.256	0.006	0.018	0.983	1.000
federated	Dir. 0.1, K=10	no DP	0.843	0.594	0.845	0.985	0.769	0.982
federated	Dir. 0.1, K=10	$\varepsilon = 8$	0.677	0.441	0.712	0.979	0.434	0.776

Mode	Partition	Regime	Bal. acc.	SARTAJ (4 cl.)	Figshare (3 cl.)	Tumour sens.	Spec. SARTAJ	Spec. Br35H
federated	Dir. 0.1, K=10	$\varepsilon = 3$	0.632	0.429	0.690	0.972	0.342	0.636
federated	Dir. 0.1, K=10	$\varepsilon = 1$	0.322	0.317	0.359	0.948	0.183	0.209
federated	Dir. 0.5, K=10	no DP	0.878	0.676	0.861	0.984	0.936	1.000
federated	Dir. 0.5, K=10	$\varepsilon = 8$	0.718	0.465	0.716	0.970	0.576	0.945
federated	Dir. 0.5, K=10	$\varepsilon = 3$	0.626	0.434	0.662	0.973	0.397	0.688
federated	Dir. 0.5, K=10	$\varepsilon = 1$	0.341	0.326	0.345	0.878	0.292	0.358
federated	Dir. 5.0, K=10	no DP	0.899	0.631	0.903	0.986	0.885	1.000
federated	Dir. 5.0, K=10	$\varepsilon = 8$	0.720	0.462	0.714	0.968	0.607	0.958
federated	Dir. 5.0, K=10	$\varepsilon = 3$	0.623	0.428	0.658	0.975	0.393	0.691
federated	Dir. 5.0, K=10	$\varepsilon = 1$	0.333	0.326	0.354	0.901	0.247	0.282
federated	IID, K=10	no DP	0.904	0.650	0.906	0.985	0.902	1.000
federated	IID, K=10	$\varepsilon = 8$	0.720	0.464	0.714	0.967	0.600	0.961
federated	IID, K=10	$\varepsilon = 3$	0.639	0.432	0.661	0.967	0.427	0.767
federated	IID, K=10	$\varepsilon = 1$	0.348	0.335	0.363	0.879	0.288	0.318
federated	natural3	no DP	0.875	0.599	0.913	0.990	0.678	0.955
federated	natural3	$\varepsilon = 8$	0.808	0.517	0.852	0.976	0.590	0.906
federated	natural3	$\varepsilon = 3$	0.714	0.498	0.716	0.977	0.593	0.891
federated	natural3	$\varepsilon = 1$	0.537	0.357	0.603	0.965	0.312	0.415
federated	natural3	patient $\varepsilon = 8$	0.403	0.312	0.503	0.967	0.092	0.152
federated	unit Dir. 0.5, K=10	patient $\varepsilon = 8$	0.264	0.255	0.184	0.497	0.549	0.479

The probe in Table S6 asks whether penultimate features reveal the source collection within one class. Trained models decode source only slightly better than an ImageNet-initialised network and a random network, so the source signal lives in the images rather than being learned during training.

Table S6: Source probe. Balanced accuracy of a logistic-regression probe on penultimate features that predicts the source collection within one class, Figshare against SARTAJ among tumour images and SARTAJ against Br35H among no-tumour images.

Mode	Partition	Regime	Tumour probe	No-tumour probe
central	natural3	no DP	0.913 (0.888, 0.938)	0.821 (0.812, 0.831)
central	natural3	clip only	0.908 (0.878, 0.938)	0.808 (0.795, 0.821)
central	natural3	$\varepsilon = 8$	0.896 (0.882, 0.909)	0.794 (0.757, 0.830)
central	natural3	$\varepsilon = 3$	0.882 (0.824, 0.941)	0.770 (0.735, 0.806)
central	natural3	$\varepsilon = 1$	0.851 (0.821, 0.882)	0.769 (0.699, 0.839)
central	natural3	patient clip only	0.914 (0.889, 0.939)	0.813 (0.791, 0.835)
central	natural3	patient $\varepsilon = 64$	0.876 (0.857, 0.895)	0.812 (0.781, 0.843)
central	natural3	patient $\varepsilon = 32$	0.864 (0.832, 0.895)	0.762 (0.730, 0.793)
central	natural3	patient $\varepsilon = 16$	0.827 (0.798, 0.855)	0.780 (0.744, 0.816)
central	natural3	patient $\varepsilon = 8$	0.828 (0.784, 0.871)	0.772 (0.747, 0.797)
central	natural3	patient $\varepsilon = 3$	0.794 (0.736, 0.851)	0.753 (0.699, 0.807)
federated	Dir. 0.1, K=10	no DP	0.920 (0.886, 0.955)	0.827 (0.814, 0.841)
federated	Dir. 0.1, K=10	$\varepsilon = 8$	0.895 (0.863, 0.927)	0.807 (0.790, 0.825)
federated	Dir. 0.1, K=10	$\varepsilon = 3$	0.891 (0.862, 0.920)	0.781 (0.758, 0.805)
federated	Dir. 0.1, K=10	$\varepsilon = 1$	0.868 (0.827, 0.909)	0.812 (0.763, 0.861)
federated	Dir. 0.5, K=10	no DP	0.930 (0.919, 0.940)	0.831 (0.814, 0.849)
federated	Dir. 0.5, K=10	$\varepsilon = 8$	0.901 (0.870, 0.932)	0.813 (0.789, 0.836)
federated	Dir. 0.5, K=10	$\varepsilon = 3$	0.891 (0.861, 0.922)	0.814 (0.765, 0.863)
federated	Dir. 0.5, K=10	$\varepsilon = 1$	0.859 (0.821, 0.898)	0.791 (0.756, 0.826)
federated	Dir. 5.0, K=10	no DP	0.921 (0.895, 0.946)	0.825 (0.804, 0.846)
federated	Dir. 5.0, K=10	$\varepsilon = 8$	0.893 (0.875, 0.911)	0.800 (0.765, 0.834)
federated	Dir. 5.0, K=10	$\varepsilon = 3$	0.893 (0.859, 0.926)	0.813 (0.758, 0.868)
federated	Dir. 5.0, K=10	$\varepsilon = 1$	0.872 (0.842, 0.901)	0.797 (0.764, 0.830)
federated	IID, K=10	no DP	0.931 (0.906, 0.955)	0.839 (0.821, 0.857)
federated	IID, K=10	$\varepsilon = 8$	0.897 (0.872, 0.921)	0.813 (0.781, 0.844)
federated	IID, K=10	$\varepsilon = 3$	0.883 (0.859, 0.907)	0.821 (0.769, 0.873)
federated	IID, K=10	$\varepsilon = 1$	0.865 (0.822, 0.909)	0.775 (0.745, 0.805)
federated	natural3	no DP	0.903 (0.890, 0.916)	0.808 (0.755, 0.861)
federated	natural3	$\varepsilon = 8$	0.871 (0.845, 0.898)	0.811 (0.782, 0.841)
federated	natural3	$\varepsilon = 3$	0.875 (0.845, 0.904)	0.799 (0.776, 0.823)
federated	natural3	$\varepsilon = 1$	0.847 (0.809, 0.886)	0.790 (0.772, 0.807)
federated	natural3	patient $\varepsilon = 8$	0.840 (0.782, 0.898)	0.799 (0.706, 0.892)
federated	unit Dir. 0.5, K=10	patient $\varepsilon = 8$	0.822 (0.769, 0.876)	0.759 (0.727, 0.790)
baseline	–	ImageNet initialisation	0.871	0.783

Mode	Partition	Regime	Tumour probe	No-tumour probe
baseline	–	random network	0.846	0.776

S3 Privacy accounting and selection cost

Table S7 gives the accounted privacy loss of every private configuration. Learning rates for federated models were chosen among two candidates trained on the same records, and composing the guarantee over both candidates raises $\varepsilon = 8$ to about 12, following the analysis of Papernot and Steinke for hyperparameter search.

Table S7: Accounted privacy loss at $\delta = 10^{-5}$ over 1000 local steps with $q = 0.1$. $\varepsilon_{k=2}$ composes the guarantee over the two learning-rate candidates trained on the same records, and the worst-unit ε composes a patient-level guarantee over the clients that hold one unit.

Mode	Partition	Unit	Target ε	σ	ε	$\varepsilon_{k=2}$	Worst-unit ε
central	natural3	image	1	12.885	1.00	–	–
central	natural3	image	3	4.832	3.00	–	–
central	natural3	patient	3	4.832	3.00	–	3.00
central	natural3	image	8	2.179	7.96	–	–
central	natural3	patient	8	2.179	7.96	–	7.96
central	natural3	patient	16	1.353	15.96	–	15.96
central	natural3	patient	32	0.922	31.90	–	31.90
central	natural3	patient	64	0.673	63.69	–	63.69
federated	Dir. 0.1, K=10	image	1	12.885	1.00	1.46	–
federated	Dir. 0.1, K=10	image	3	4.832	3.00	4.42	–
federated	Dir. 0.1, K=10	image	8	2.179	7.96	11.98	–
federated	Dir. 0.5, K=10	image	1	12.885	1.00	1.46	–
federated	Dir. 0.5, K=10	image	3	4.832	3.00	4.42	–
federated	Dir. 0.5, K=10	image	8	2.179	7.96	11.98	–
federated	Dir. 0.5, K=15	image	3	4.832	3.00	4.42	–
federated	Dir. 0.5, K=15	image	8	2.179	7.96	11.98	–
federated	Dir. 0.5, K=3	image	3	4.832	3.00	4.42	–
federated	Dir. 0.5, K=3	image	8	2.179	7.96	11.98	–
federated	Dir. 0.5, K=5	image	3	4.832	3.00	4.42	–
federated	Dir. 0.5, K=5	image	8	2.179	7.96	11.98	–
federated	Dir. 5.0, K=10	image	1	12.885	1.00	1.46	–
federated	Dir. 5.0, K=10	image	3	4.832	3.00	4.42	–
federated	Dir. 5.0, K=10	image	8	2.179	7.96	11.98	–
federated	IID, K=10	image	1	12.885	1.00	1.46	–
federated	IID, K=10	image	3	4.832	3.00	4.42	–
federated	IID, K=10	image	8	2.179	7.96	11.98	–
federated	natural3	image	1	12.885	1.00	1.46	–
federated	natural3	image	3	4.832	3.00	4.42	–
federated	natural3	image	8	2.179	7.96	11.98	–
federated	natural3	patient	8	2.179	7.96	18.39	11.98
federated	unit Dir. 0.5, K=10	image	3	4.832	3.00	4.42	–
federated	unit Dir. 0.5, K=10	image	8	2.179	7.96	11.98	–
federated	unit Dir. 0.5, K=10	patient	8	2.179	7.96	11.98	7.96

Central models use the learning rate 0.02 fixed in pilot runs on the same training data, so their selection cost is disclosed but not composed. Natural3 splits 11 units across two clients, which raises the worst-unit guarantee of that partition.

Table S8 lists the learning-rate candidates at seed 0 and the one selected by lowest validation loss. Every non-private configuration selected 0.01, while private ones mostly selected 0.003, which reflects the smaller steps that noisy gradients tolerate.

Table S8: Learning-rate selection at seed 0 for federated models, by lowest best validation loss among two candidates. Seed 0 is never used for reported results.

Partition	Regime	lr	Best val. loss	Val. bal. acc.	Selected
natural3	no DP	0.003	0.447	0.801	
natural3	no DP	0.01	0.333	0.847	✓

Partition	Regime	lr	Best val. loss	Val. bal. acc.	Selected
IID, K=10	no DP	0.003	0.310	0.882	
IID, K=10	no DP	0.01	0.255	0.911	✓
Dir. 0.1, K=10	no DP	0.003	0.542	0.720	
Dir. 0.1, K=10	no DP	0.01	0.378	0.838	✓
Dir. 0.5, K=10	no DP	0.003	0.356	0.836	
Dir. 0.5, K=10	no DP	0.01	0.318	0.889	✓
Dir. 5.0, K=10	no DP	0.003	0.309	0.875	
Dir. 5.0, K=10	no DP	0.01	0.267	0.896	✓
natural3	$\varepsilon = 8$	0.003	0.780	0.678	
natural3	$\varepsilon = 8$	0.01	0.656	0.811	✓
IID, K=10	$\varepsilon = 8$	0.003	0.784	0.691	✓
IID, K=10	$\varepsilon = 8$	0.01	0.823	0.648	
Dir. 0.1, K=10	$\varepsilon = 8$	0.003	0.751	0.689	✓
Dir. 0.1, K=10	$\varepsilon = 8$	0.01	0.901	0.568	
Dir. 0.5, K=10	$\varepsilon = 8$	0.003	0.763	0.692	✓
Dir. 0.5, K=10	$\varepsilon = 8$	0.01	0.877	0.644	
Dir. 5.0, K=10	$\varepsilon = 8$	0.003	0.779	0.685	✓
Dir. 5.0, K=10	$\varepsilon = 8$	0.01	0.835	0.625	
natural3	$\varepsilon = 3$	0.003	0.784	0.663	✓
natural3	$\varepsilon = 3$	0.01	0.812	0.588	
IID, K=10	$\varepsilon = 3$	0.003	0.928	0.504	✓
IID, K=10	$\varepsilon = 3$	0.01	1.228	0.336	
Dir. 0.1, K=10	$\varepsilon = 3$	0.003	0.957	0.521	✓
Dir. 0.1, K=10	$\varepsilon = 3$	0.01	1.241	0.299	
Dir. 0.5, K=10	$\varepsilon = 3$	0.003	0.970	0.560	✓
Dir. 0.5, K=10	$\varepsilon = 3$	0.01	1.255	0.299	
Dir. 5.0, K=10	$\varepsilon = 3$	0.003	0.953	0.483	✓
Dir. 5.0, K=10	$\varepsilon = 3$	0.01	1.233	0.308	
natural3	$\varepsilon = 1$	0.003	0.980	0.518	✓
natural3	$\varepsilon = 1$	0.01	1.363	0.274	
IID, K=10	$\varepsilon = 1$	0.003	1.307	0.389	✓
IID, K=10	$\varepsilon = 1$	0.01	1.769	0.275	
Dir. 0.1, K=10	$\varepsilon = 1$	0.003	1.243	0.372	✓
Dir. 0.1, K=10	$\varepsilon = 1$	0.01	1.889	0.251	
Dir. 0.5, K=10	$\varepsilon = 1$	0.003	1.311	0.384	✓
Dir. 0.5, K=10	$\varepsilon = 1$	0.01	1.738	0.269	
Dir. 5.0, K=10	$\varepsilon = 1$	0.003	1.311	0.380	✓
Dir. 5.0, K=10	$\varepsilon = 1$	0.01	1.693	0.276	
natural3	patient $\varepsilon = 8$	0.003	1.191	0.408	✓
natural3	patient $\varepsilon = 8$	0.01	1.543	0.280	
unit Dir. 0.5, K=10	patient $\varepsilon = 8$	0.003	2.098	0.353	
unit Dir. 0.5, K=10	patient $\varepsilon = 8$	0.01	2.051	0.246	✓
natural3	patient $\varepsilon = 3$	0.003	1.410	0.267	✓
natural3	patient $\varepsilon = 3$	0.01	15.649	0.281	
unit Dir. 0.5, K=10	patient $\varepsilon = 3$	0.003	2.275	0.214	✓
unit Dir. 0.5, K=10	patient $\varepsilon = 3$	0.01	15.845	0.250	
unit Dir. 0.5, K=10	no DP	0.003	0.385	0.843	
unit Dir. 0.5, K=10	no DP	0.01	0.306	0.886	✓
unit Dir. 0.5, K=10	$\varepsilon = 8$	0.003	0.783	0.681	✓
unit Dir. 0.5, K=10	$\varepsilon = 8$	0.01	0.911	0.520	
unit Dir. 0.5, K=10	$\varepsilon = 3$	0.003	0.958	0.599	✓
unit Dir. 0.5, K=10	$\varepsilon = 3$	0.01	1.292	0.322	

S4 Client count and drift correction

Table S9 holds the client-count experiment at the shared learning rate of 0.003. The aggregated noise standard deviation grows with $\sqrt{K}$ at a fixed noise multiplier, and accuracy at $\varepsilon = 3$ falls as it grows, whereas accuracy without privacy shows no trend.

Table S9: Client count at fixed data, Dirichlet $\alpha = 0.5$ FedAvg at learning rate 0.003, full test set. The aggregated noise standard deviation is $\sigma C\sqrt{K}/(qN)$ with $N = 2610$.

K	Regime	Client size min/med./max	σ	Agg. noise s.d.	Bal. acc.	sens$_M$
3	$\varepsilon = 3$	424/618/1568	4.832	0.0321	0.719 (0.698, 0.739)	0.315
5	$\varepsilon = 3$	204/452/1065	4.832	0.0414	0.708 (0.684, 0.732)	0.171
10	$\varepsilon = 3$	82/225/533	4.832	0.0585	0.626 (0.570, 0.682)	0.067
15	$\varepsilon = 3$	29/190/354	4.832	0.0717	0.558 (0.458, 0.658)	0.030
3	$\varepsilon = 8$	424/618/1568	2.179	0.0145	0.717 (0.644, 0.790)	0.305
5	$\varepsilon = 8$	204/452/1065	2.179	0.0187	0.735 (0.712, 0.758)	0.244
10	$\varepsilon = 8$	82/225/533	2.179	0.0264	0.718 (0.686, 0.750)	0.193
15	$\varepsilon = 8$	29/190/354	2.179	0.0323	0.696 (0.618, 0.774)	0.163
3	no DP	424/618/1568	0.000	0.0000	0.868 (0.850, 0.886)	0.643
5	no DP	204/452/1065	0.000	0.0000	0.906 (0.894, 0.919)	0.811
10	no DP	82/225/533	0.000	0.0000	0.878 (0.863, 0.893)	0.605
15	no DP	29/190/354	0.000	0.0000	0.903 (0.897, 0.909)	0.799

FedProx screening at seed 0 (Table S10) shows that proximal weights up to 0.1 change validation loss only in the third decimal place. Larger weights helped in a few private settings and hurt elsewhere, without a consistent pattern that would justify carrying FedProx to the seed runs.

Table S10: FedProx screening at seed 0. Best validation loss of FedAvg ($\mu = 0$) and of FedProx with proximal weight μ, at the learning rate selected for FedAvg.

Partition	Regime	lr	$\mu = 0$	$\mu = 0.01$	$\mu = 0.1$	$\mu = 1$	$\mu = 10$
Dir. 0.1, K=10	$\varepsilon = 3$	0.003	0.957	0.957	0.954	0.944	0.930
Dir. 0.1, K=10	$\varepsilon = 8$	0.003	0.751	0.751	0.751	0.751	0.770
Dir. 0.1, K=10	no DP	0.01	0.378	0.374	0.378	0.405	0.570
Dir. 0.5, K=10	$\varepsilon = 3$	0.003	0.970	0.970	0.969	0.962	0.906
Dir. 0.5, K=10	$\varepsilon = 8$	0.003	0.763	0.763	0.764	0.767	0.812
Dir. 0.5, K=10	no DP	0.01	0.318	0.318	0.317	0.323	0.339
Dir. 5.0, K=10	$\varepsilon = 3$	0.003	0.953	0.953	0.952	0.942	0.911
Dir. 5.0, K=10	$\varepsilon = 8$	0.003	0.779	0.779	0.779	0.783	0.819
Dir. 5.0, K=10	no DP	0.01	0.267	0.267	0.266	0.266	0.300
IID, K=10	$\varepsilon = 3$	0.003	0.928	0.928	0.928	0.924	0.900
IID, K=10	$\varepsilon = 8$	0.003	0.784	0.784	0.784	0.786	0.823
IID, K=10	no DP	0.01	0.255	0.255	0.254	0.262	0.290
natural3	$\varepsilon = 3$	0.003	0.784	0.784	0.785	0.789	0.838
natural3	$\varepsilon = 8$	0.01	0.656	0.658	0.656	0.665	0.734
natural3	no DP	0.01	0.333	0.340	0.343	0.371	0.516

Table S11 compares size-weighted SCAFFOLD with FedAvg. No difference survives Holm correction, and the uniform server average of the original algorithm collapses on unequal clients under privacy, most visibly on natural3, where one client holds 78% of the images.

Table S11: SCAFFOLD against FedAvg on the full test set. Size-weighted SCAFFOLD over seeds 1 to 5 with paired t-test and Holm correction across 15 configurations, and the uniform server average of the original algorithm over seeds 1 to 3.

Partition	Regime	FedAvg	SCAFFOLD (size)	Diff.	p	p Holm	SCAFFOLD (uniform)
natural3	no DP	0.875	0.900	+0.025	0.0043	0.064	0.898
Dir. 5.0, K=10	$\varepsilon = 8$	0.720	0.716	-0.004	0.0054	0.075	0.713
Dir. 0.1, K=10	no DP	0.843	0.815	-0.028	0.0132	0.172	0.761
natural3	$\varepsilon = 8$	0.808	0.825	+0.017	0.0136	0.172	0.590
Dir. 0.5, K=10	no DP	0.878	0.889	+0.011	0.0514	0.566	0.886
Dir. 0.1, K=10	$\varepsilon = 8$	0.677	0.705	+0.028	0.1132	1.000	0.428
IID, K=10	no DP	0.904	0.909	+0.005	0.2159	1.000	0.904
IID, K=10	$\varepsilon = 3$	0.639	0.637	-0.003	0.2306	1.000	0.642
Dir. 0.5, K=10	$\varepsilon = 3$	0.626	0.621	-0.006	0.3204	1.000	0.492
Dir. 5.0, K=10	no DP	0.898	0.907	+0.008	0.3871	1.000	0.903
Dir. 0.5, K=10	$\varepsilon = 8$	0.718	0.713	-0.005	0.4677	1.000	0.718
IID, K=10	$\varepsilon = 8$	0.720	0.719	-0.001	0.5066	1.000	0.728
natural3	$\varepsilon = 3$	0.714	0.713	-0.001	0.7524	1.000	0.358
Dir. 5.0, K=10	$\varepsilon = 3$	0.623	0.625	+0.001	0.8260	1.000	0.642
Dir. 0.1, K=10	$\varepsilon = 3$	0.632	0.633	+0.001	0.9456	1.000	0.268

S5 Membership inference

The weight-noise comparator used a rule fixed before any seed run. Table S12 shows that the literal $s = 0.5$ of the prior study diverged, and the rule selected $s = 0.003$ as the largest level whose validation loss did not exceed that of record-level $\varepsilon = 8$.

Table S12: Selection of the utility-matched weight-noise level at seed 0 on natural3. Gaussian noise $\mathcal{N}(0, s^2)$ is added to each client's weights after local training, without clipping.

s	Diverged	Best val. loss	Meets reference
0.5	yes	21.214	no
0.05	no	15.452	no
0.01	no	0.945	no
0.003	no	0.357	yes
0.001	no	0.313	yes

The reference is the seed-0 validation loss of record-level $\varepsilon = 8$ FedAvg on natural3, 0.656. The rule takes the largest s that meets it. Over seeds 1 to 5 on the full test set, weight noise $s = 0.5$ reaches balanced accuracy 0.259 (0.235, 0.284) and weight noise $s = 0.003$ reaches balanced accuracy 0.843 (0.826, 0.860).

Tables S13 and S14 report LiRA with the clean and with the full test set as non-members, in both its online and its offline form. The choice of non-member set changes no conclusion, since online AUC, true-positive rate at 1% false positives and patient AUC move by less than 0.01 for every target that produced a usable model, and every private setting stays at the level of a random guess under both definitions.

Table S13: LiRA against natural3 FedAvg with 16 shadow models, clean test set (747 images) as non-members. Online and offline variants, mean over target seeds with 95% t interval.

Target	n	AUC	TPR@1%	TPR@0.1%	Patient AUC	Off. AUC	Off. TPR@1%	Off. patient AUC
no DP	5	0.666 (0.642, 0.689)	0.108 (0.081, 0.135)	0.033 (0.019, 0.046)	0.805 (0.762, 0.848)	0.656	0.068	0.769
$\varepsilon = 8$	5	0.464 (0.448, 0.480)	0.017 (0.012, 0.022)	0.006 (0.003, 0.008)	0.454 (0.427, 0.481)	0.503	0.016	0.511
$\varepsilon = 3$	5	0.545 (0.522, 0.567)	0.015 (0.008, 0.023)	0.003 (0.002, 0.005)	0.567 (0.529, 0.605)	0.546	0.014	0.581
$\varepsilon = 1$	5	0.479 (0.469, 0.489)	0.010 (0.006, 0.015)	0.003 (0.000, 0.005)	0.457 (0.413, 0.500)	0.488	0.012	0.500
weight noise $s = 0.5$	5	0.474 (0.437, 0.511)	0.038 (0.023, 0.053)	0.003 (0.002, 0.003)	0.518 (0.497, 0.539)	0.538	0.034	0.545
weight noise $s = 0.003$	5	0.594 (0.572, 0.616)	0.045 (0.038, 0.051)	0.016 (0.012, 0.020)	0.745 (0.713, 0.778)	0.569	0.030	0.665

Table S14: LiRA against natural3 FedAvg with 16 shadow models, full test set (813 images) as non-members. Online and offline variants, mean over target seeds with 95% t interval.

Target	n	AUC	TPR@1%	TPR@0.1%	Patient AUC	Off. AUC	Off. TPR@1%	Off. patient AUC
no DP	5	0.662 (0.642, 0.682)	0.112 (0.084, 0.140)	0.033 (0.019, 0.046)	0.805 (0.762, 0.848)	0.661	0.069	0.769
$\varepsilon = 8$	5	0.472 (0.459, 0.484)	0.015 (0.010, 0.020)	0.002 (-0.000, 0.005)	0.454 (0.427, 0.481)	0.497	0.013	0.511
$\varepsilon = 3$	5	0.545 (0.524, 0.565)	0.017 (0.010, 0.023)	0.003 (0.002, 0.005)	0.567 (0.529, 0.604)	0.546	0.014	0.581
$\varepsilon = 1$	5	0.483 (0.473, 0.493)	0.009 (0.005, 0.013)	0.001 (-0.000, 0.002)	0.456 (0.412, 0.500)	0.489	0.010	0.499
weight noise $s = 0.5$	5	0.480 (0.455, 0.504)	0.022 (0.014, 0.030)	0.003 (0.002, 0.003)	0.518 (0.497, 0.538)	0.529	0.024	0.545
weight noise $s = 0.003$	5	0.589 (0.568, 0.610)	0.047 (0.038, 0.055)	0.016 (0.012, 0.020)	0.745 (0.714, 0.777)	0.585	0.033	0.665

The loss-threshold attack in Table S15 is far weaker than LiRA even without privacy. Its values below 0.5 under privacy persist when the attack is computed within each source and class stratum, so they do not arise from the mix of sources and classes in each split.

Table S15: Loss-threshold membership attack, AUC of the negative per-example loss with training images as members and test images as non-members. Within-stratum AUC averages the AUC computed inside each source and class stratum.

Mode	Partition	Regime	Pooled (full)	Stratum (full)	Pooled (clean)	Stratum (clean)	Figshare (clean)
central	natural3	no DP	0.540	0.531	0.550	0.533	0.540
central	natural3	clip only	0.494	0.479	0.510	0.478	0.488
central	natural3	$\varepsilon = 8$	0.485	0.466	0.497	0.465	0.477
central	natural3	$\varepsilon = 3$	0.493	0.474	0.499	0.474	0.492
central	natural3	$\varepsilon = 1$	0.513	0.500	0.498	0.499	0.517
central	natural3	patient $\varepsilon = 8$	0.511	0.507	0.546	0.507	0.505
central	natural3	patient $\varepsilon = 3$	0.506	0.502	0.543	0.502	0.499
central	natural3	patient clip only	0.505	0.477	0.539	0.477	0.482
central	natural3	patient $\varepsilon = 16$	0.495	0.478	0.529	0.477	0.481
central	natural3	patient $\varepsilon = 32$	0.501	0.481	0.536	0.481	0.488
central	natural3	patient $\varepsilon = 64$	0.495	0.471	0.529	0.470	0.472
federated	Dir. 0.1, K=10	$\varepsilon = 1$	0.505	0.503	0.496	0.503	0.514
federated	Dir. 0.1, K=10	$\varepsilon = 3$	0.493	0.480	0.485	0.480	0.491
federated	Dir. 0.1, K=10	$\varepsilon = 8$	0.496	0.482	0.494	0.482	0.495
federated	Dir. 0.1, K=10	no DP	0.518	0.506	0.527	0.506	0.515
federated	Dir. 0.5, K=10	$\varepsilon = 1$	0.508	0.496	0.504	0.496	0.515
federated	Dir. 0.5, K=10	$\varepsilon = 3$	0.501	0.491	0.495	0.491	0.502
federated	Dir. 0.5, K=10	$\varepsilon = 8$	0.496	0.483	0.501	0.482	0.495
federated	Dir. 0.5, K=10	no DP	0.526	0.518	0.548	0.519	0.521
federated	Dir. 5.0, K=10	$\varepsilon = 1$	0.507	0.496	0.499	0.496	0.515
federated	Dir. 5.0, K=10	$\varepsilon = 3$	0.501	0.491	0.495	0.491	0.502
federated	Dir. 5.0, K=10	$\varepsilon = 8$	0.495	0.479	0.501	0.479	0.492
federated	Dir. 5.0, K=10	no DP	0.526	0.521	0.537	0.522	0.526
federated	IID, K=10	$\varepsilon = 1$	0.507	0.493	0.502	0.493	0.515
federated	IID, K=10	$\varepsilon = 3$	0.498	0.486	0.495	0.487	0.500
federated	IID, K=10	$\varepsilon = 8$	0.495	0.479	0.501	0.479	0.493
federated	IID, K=10	no DP	0.532	0.528	0.546	0.529	0.533
federated	natural3	$\varepsilon = 1$	0.497	0.475	0.487	0.474	0.498
federated	natural3	$\varepsilon = 3$	0.490	0.473	0.491	0.472	0.487
federated	natural3	$\varepsilon = 8$	0.476	0.459	0.478	0.458	0.470
federated	natural3	no DP	0.532	0.526	0.521	0.527	0.535
federated	natural3	patient $\varepsilon = 8$	0.499	0.489	0.477	0.488	0.502
federated	unit Dir. 0.5, K=10	patient $\varepsilon = 8$	0.497	0.498	0.519	0.496	0.488
federated	natural3	weight noise $s = 0.5$	0.506	0.497	0.487	0.497	0.514
federated	natural3	weight noise $s = 0.003$	0.505	0.498	0.488	0.499	0.504

S6 Explanations

Table S16 reports localisation and faithfulness for Grad-CAM, Grad-CAM++ and Integrated Gradients across the privacy settings of central training. Grad-CAM shows the clearest decline as the privacy budget shrinks, Grad-CAM++ declines more gently, and Integrated Gradients reaches the highest pointing rate without privacy, a result that the randomisation check below qualifies.

Table S16: Localisation and faithfulness of three attribution methods for central models on correctly classified Figshare test tumour images with expert masks, seed means. Lower deletion and higher insertion AUC indicate more faithful maps.

Regime	Method	Pointing	IoU top 15%	Energy in mask	Deletion	Insertion
no DP	Grad-CAM	0.197	0.087	0.032	0.290	0.604
no DP	Grad-CAM++	0.139	0.080	0.032	0.310	0.585
no DP	Int. Gradients	0.276	0.058	0.065	0.151	0.462
clip only	Grad-CAM	0.190	0.086	0.031	0.382	0.689
clip only	Grad-CAM++	0.152	0.084	0.032	0.389	0.676
clip only	Int. Gradients	0.200	0.054	0.057	0.214	0.479
$\varepsilon = 8$	Grad-CAM	0.156	0.076	0.029	0.406	0.715
$\varepsilon = 8$	Grad-CAM++	0.129	0.079	0.030	0.429	0.705
$\varepsilon = 8$	Int. Gradients	0.198	0.051	0.053	0.311	0.510
$\varepsilon = 3$	Grad-CAM	0.103	0.055	0.027	0.490	0.688
$\varepsilon = 3$	Grad-CAM++	0.098	0.069	0.028	0.494	0.708
$\varepsilon = 3$	Int. Gradients	0.172	0.046	0.048	0.353	0.531
$\varepsilon = 1$	Grad-CAM	0.089	0.032	0.027	0.480	0.526
$\varepsilon = 1$	Grad-CAM++	0.123	0.065	0.031	0.469	0.546
$\varepsilon = 1$	Int. Gradients	0.123	0.047	0.049	0.469	0.595
patient clip only	Grad-CAM	0.187	0.085	0.032	0.307	0.620
patient clip only	Grad-CAM++	0.115	0.077	0.032	0.330	0.590
patient clip only	Int. Gradients	0.239	0.055	0.060	0.155	0.396
patient $\varepsilon = 64$	Grad-CAM	0.127	0.058	0.029	0.290	0.555
patient $\varepsilon = 64$	Grad-CAM++	0.147	0.079	0.031	0.285	0.568
patient $\varepsilon = 64$	Int. Gradients	0.193	0.052	0.054	0.103	0.363
patient $\varepsilon = 32$	Grad-CAM	0.097	0.042	0.027	0.267	0.459
patient $\varepsilon = 32$	Grad-CAM++	0.129	0.075	0.029	0.251	0.521
patient $\varepsilon = 32$	Int. Gradients	0.168	0.047	0.048	0.102	0.322
patient $\varepsilon = 16$	Grad-CAM	0.069	0.032	0.025	0.269	0.375
patient $\varepsilon = 16$	Grad-CAM++	0.111	0.063	0.028	0.252	0.433
patient $\varepsilon = 16$	Int. Gradients	0.147	0.046	0.047	0.105	0.247
patient $\varepsilon = 8$	Grad-CAM	0.082	0.029	0.025	0.276	0.314
patient $\varepsilon = 8$	Grad-CAM++	0.102	0.047	0.028	0.238	0.363
patient $\varepsilon = 8$	Int. Gradients	0.206	0.049	0.051	0.163	0.248
patient $\varepsilon = 3$	Grad-CAM	0.056	0.023	0.018	0.345	0.300
patient $\varepsilon = 3$	Grad-CAM++	0.000	0.030	0.018	0.281	0.334
patient $\varepsilon = 3$	Int. Gradients	0.000	0.039	0.039	0.272	0.162

The randomisation check in Table S17 decides which method deserves interpretation. Integrated Gradients with a black baseline keeps a rank correlation near 0.7 with maps from a network whose weights were all re-initialised, so its maps mostly reflect the input rather than the learned model, whereas Grad-CAM correlations stay between -0.05 and 0.39.

Table S17: Model-parameter randomisation check, Spearman rank correlation between maps of the trained model and of a fully re-initialised network (central models, seed 1). Values near zero pass the check. All layers re-initialises every weight, last block only the final residual block and the classifier.

Regime	Grad-CAM, all layers	Grad-CAM, last block	IG, all layers	IG, last block
no DP	0.310	0.444	0.684	0.709
clip only	0.328	0.408	0.700	0.709
$\varepsilon = 8$	0.388	0.377	0.706	0.714
$\varepsilon = 3$	0.324	0.029	0.699	0.722
$\varepsilon = 1$	-0.047	0.309	0.703	0.728
patient clip only	0.166	0.477	0.683	0.705
patient $\varepsilon = 64$	0.232	0.491	0.694	0.719
patient $\varepsilon = 32$	0.309	0.403	0.703	0.726
patient $\varepsilon = 16$	0.035	-0.010	0.690	0.710
patient $\varepsilon = 8$	0.050	0.316	0.698	0.726
patient $\varepsilon = 3$	0.101	0.098	0.712	0.731

Table S18 tests for skull-border and background artefacts, adds Grad-CAM from the third residual block at twice the resolution and lists the cross-seed consistency used in the article. Grad-CAM places less mass on the brain rim and outside the brain than their area shares, so neither artefact explains the localisation results.

Table S18: Border-rim analysis, layer-3 Grad-CAM (8×8) and cross-seed consistency. Rim and outside-brain values are the share of Grad-CAM mass, next to the share of image area (chance).

Mode	Partition	Regime	Rim	Rim chance	Outside	Outside chance	L3 pointing	L3 IoU	Consistency
central	natural3	no DP	0.175	0.334	0.261	0.405	0.192	0.055	0.700
central	natural3	clip only	0.174	0.334	0.266	0.401	0.142	0.043	0.663
central	natural3	$\varepsilon = 8$	0.176	0.334	0.280	0.398	0.123	0.044	0.485
central	natural3	$\varepsilon = 3$	0.180	0.333	0.294	0.389	0.103	0.038	0.309
central	natural3	$\varepsilon = 1$	0.180	0.351	0.362	0.443	0.051	0.027	0.235
central	natural3	patient $\varepsilon = 8$	0.180	0.332	0.333	0.417	0.060	0.028	0.218
central	natural3	patient $\varepsilon = 3$	0.176	0.308	0.289	0.329	0.056	0.009	0.234
central	natural3	patient clip only	0.175	0.335	0.280	0.405	0.145	0.048	0.609
central	natural3	patient $\varepsilon = 16$	0.181	0.335	0.318	0.406	0.066	0.030	0.225
central	natural3	patient $\varepsilon = 32$	0.176	0.332	0.296	0.395	0.113	0.040	0.291
central	natural3	patient $\varepsilon = 64$	0.174	0.332	0.299	0.399	0.156	0.049	0.368
federated	Dir. 0.1, K=10	$\varepsilon = 1$	0.163	0.358	0.406	0.452	0.056	0.024	0.291
federated	Dir. 0.1, K=10	$\varepsilon = 3$	0.176	0.335	0.290	0.395	0.045	0.021	0.364
federated	Dir. 0.1, K=10	$\varepsilon = 8$	0.174	0.334	0.274	0.395	0.120	0.037	0.489
federated	Dir. 0.1, K=10	no DP	0.173	0.334	0.269	0.399	0.101	0.037	0.687
federated	Dir. 0.5, K=10	$\varepsilon = 1$	0.193	0.359	0.407	0.462	0.051	0.026	0.302
federated	Dir. 0.5, K=10	$\varepsilon = 3$	0.176	0.331	0.258	0.383	0.082	0.035	0.451
federated	Dir. 0.5, K=10	$\varepsilon = 8$	0.173	0.333	0.258	0.390	0.131	0.043	0.513
federated	Dir. 0.5, K=10	no DP	0.174	0.335	0.261	0.401	0.161	0.049	0.725
federated	Dir. 5.0, K=10	$\varepsilon = 1$	0.188	0.359	0.406	0.458	0.049	0.022	0.312
federated	Dir. 5.0, K=10	$\varepsilon = 3$	0.176	0.330	0.258	0.382	0.088	0.034	0.456
federated	Dir. 5.0, K=10	$\varepsilon = 8$	0.173	0.333	0.255	0.389	0.141	0.043	0.523
federated	Dir. 5.0, K=10	no DP	0.173	0.335	0.264	0.404	0.157	0.048	0.703
federated	IID, K=10	$\varepsilon = 1$	0.192	0.360	0.416	0.462	0.048	0.022	0.303
federated	IID, K=10	$\varepsilon = 3$	0.175	0.330	0.251	0.381	0.089	0.038	0.476
federated	IID, K=10	$\varepsilon = 8$	0.173	0.332	0.254	0.388	0.140	0.043	0.526
federated	IID, K=10	no DP	0.173	0.334	0.262	0.403	0.155	0.048	0.705
federated	natural3	$\varepsilon = 1$	0.174	0.335	0.287	0.401	0.074	0.032	0.356
federated	natural3	$\varepsilon = 3$	0.175	0.331	0.251	0.386	0.124	0.044	0.532
federated	natural3	$\varepsilon = 8$	0.175	0.333	0.280	0.399	0.118	0.041	0.467
federated	natural3	no DP	0.174	0.335	0.267	0.405	0.118	0.041	0.657

Open questions and risks

1. The patient-level guarantee is exact only for Figshare, because SARTAJ and Br35H units are groups of similar images rather than verified patients, and some of those groups may still combine different people.

2. Mirrored pairs remain between training and validation data (47 pairs at NCC 0.95), so round and learning-rate selection drew on slightly contaminated validation data even though every test result excludes contaminated images.

3. Central models use a learning rate fixed in pilot runs on the same training data, and this selection cost is disclosed rather than composed into ε.

4. The H5 mechanism table, the SCAFFOLD comparison and the source controls are computed on the full test set, and the clean-set versions of the corresponding main-text values come from Table S3.

5. Randomisation was checked on one seed per setting with 64 images, which suffices to separate the two methods but not to rank privacy settings against each other.